

Customer Shopping Behavior Analysis

1. Project Overview:

We are analyzing 3,900 transactional data of purchases across various product categories, to analyze customer shopping behavior.

The goal is to reveal spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. About DATASET:

- Rows: 3900
- Columns: 18
- Key Features:
 - Customer demographics
 - Purchase Details
 - Shopping Behavior
- Missing Dates: 37 Values in Review Rating Column

3. Data Analysis Using Python:

Begin with Data preparation, cleaning and restructuring the data in Python.

- **Data Loading** : Importing the dataset using pandas
- **Initial Exploration**: `df.head()` to view a data sample.

```
# Loading the dataset using pandas
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')

df.head()
```

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchase	
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Vermont	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

- **`df.info()`** : To check structure of the table
- **Missing Data Handling**: Checked for null values and imputed missing values in the 'Review_Rating' column using the median rating of each product category.
- **Column Standardization**: Renamed columns to snake case for better documentation and for easy handling while analysis.
- **Creating Columns**:
 - Created an 'age_group' column.
 - Created a 'purchase_frequency_days' column.

- **Database Integration** : Connected Python script to PostgreSQL and loaded the cleaned 'Dataset' into the database for SQL analysis.

4. Data Analysis Using SQL.

I performed structured analysis in PostgreSQL to answer some of the key business analytical questions.

1. Comparison of Total revenue done by male vs done by female:

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. Identifying users who spent more than the average 'purchase_amount' even after using the discounts.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
Total rows: 839		Query complete 00:00:00.333

3. Products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Comparing average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Comparing average spend and total revenue between people with and without subscription status.

	subscription_status text	total_customer bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Identifying Products with the highest percentage of discount purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00
6	Boots	46.00
7	Jeans	45.00
8	Dress	45.00
9	Hoodie	45.00
10	Belt	44.00

7. Classifying customers into New, Returning and Loyal segments based on the customer's purchase history.

	customer_segment text 	Number of Customers bigint 
1	Loyal	3116
2	New	83
3	Returning	701

- 8.